

Feasibility Studies: Atomic Hydrogen Target

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Potentially very attractive is replacing the ferromagnetic target by ultra-cold atomic hydrogen gas, contained in a magnetic trap. Electrons in such a gas can be 100% polarized and a density of about $3 \cdot 10^{15}$ atoms/cm³ can be achieved. For the effective target length of 10 cm the target density would be $3 \cdot 10^{16}$ e⁻/cm². The currently used targets in Hall A are 30 μm thick along the beam, have a polarization of 8% and are used at 0.3 μA of beam current. A statistical accuracy of 1% for the asymmetry is obtained typically in 2 minutes. For measurement with the hydrogen target discussed let us make assumptions as follows:

- Target density $3 \cdot 10^{16}$ e⁻/cm²
- Target polarization 100%
- Beam current 100 μA
- Polarimeter acceptance is the same as for the ferromagnetic targets (MC simulation supports this assumption).

In this case one would need about 8 minutes to reach the 1% statistical accuracy. This is still a reasonably short period. Also, since the target is very thin, one might be able to run polarimetry along with the main experiment.

Usage of such a target eliminates several systematic errors:

- Target polarization, about 3%
- Levchuk effect, can be about 1%
- Target angle 0.5%
- Beam current dependence, can be 1-2%, caused by dilutions from other halls while running polarimetry at low currents
- Dead time, 0.3%
- Background from radiative Mott scattering on iron nuclei

One may hope to reduce the relative systematic error on the asymmetry to about 0.3-0.5%.

Atomic Hydrogen

Some references:

- [WWW page from University of Amsterdam](#)

Polarimeter Optics

The magnetic trap is created by a solenoid with a field of about 5 T, directed along the beam. Such a field can distort the polarimeter optics. Indeed, at 0.850 GeV the scattering plane is rotated by 4° on average, depending on the longitudinal coordinate of the interaction point. Nevertheless, the acceptance loss is small even at the lowest beam energy of 0.850 GeV. Events, simulated at 0.850 GeV are shown on the following pictures (0.85 GeV solenoid note, appended).



